

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.23	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.9	Product Info 2	A1
Occupation	WORKER	Annual Income	\$305,000.00
Insurance Type	Life	Coverage Amount	\$4,500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Life Insurance Application Review

Date: October 26, 2023

Applicant ID: [Not Provided, referring to Current Applicant]

Insurance Type: Life Insurance

Coverage Amount: \$4,500,000.00

1. Risk Assessment

The applicant presents a mixed risk profile, with several favorable factors offset by critical areas requiring clarification and attention:

* **Age:** The reported age of **0.23 years** is a significant anomaly. This suggests a data entry error. Assuming this to be 23 years (0.23 being a typo for 23.0), the applicant would be a young adult, generally associated with lower mortality risk. However, this discrepancy must be verified. If genuinely 0.23 years (an infant), the application with such high income and coverage would be highly unusual and requires immediate scrutiny.

* **BMI:** At 22.9, the applicant's BMI (Height: 175.0 cm, Weight: 70.0 kg) falls within the healthy range, indicating good physical condition in this aspect.

* **Smoking Status:** The applicant is a **smoker (1)**, which is a significant adverse risk factor for life insurance, increasing mortality and morbidity risks considerably.

* **Occupation:** 'WORKER' is a very general classification. While income of \$305,000.00 suggests a skilled or professional role, a specific occupation code would be necessary to assess occupational hazards accurately.

* **Income:** A high income of \$305,000.00 supports the financial justification for a substantial coverage amount, though the coverage multiple is high.

* **Previous Claims & Family History:** No previous claims (0) and no adverse family history (0) are positive indicators, suggesting a clean claims record and generally good genetic health predisposition.

* **Lifestyle:** Reported '0' for both exercise and alcohol consumption. While '0' for alcohol is favorable, '0' for exercise might imply a lack of regular physical activity, which could be a minor health concern over time.

2. Comparative Analysis

The current application has been assigned Risk Class 1, with a predicted annual premium of \$10,000.00. Comparing it to historical approved cases:

* **Application APP-0642B5 (Similarity: 81.0%):** This historical case is the most directly comparable for pricing, as it also received a **Risk Class 1** and was approved with an identical **Premium of \$10,000.00**. Despite a lower similarity score, this provides strong precedent for the current premium and risk classification if the underlying risk factors are truly comparable. The key unknown here is whether APP-0642B5 was also a smoker; if not, the current applicant being a smoker and still achieving Risk Class 1 warrants further investigation into the model's feature weighting.

* **Application POL-8902 (Similarity: 98.8%):** This case has very high similarity, but was assigned a **Risk Class 2** with a higher premium of \$12,000.00. The current applicant being classified as Risk Class 1 (better) suggests that despite high overall similarity, the model identified specific factors in the current application that

place it in a superior risk category compared to POL-8902. Again, the unknown smoking status of POL-8902 is critical for a full comparison.

* **Application POL-4519 (Similarity: 96.0%):** This case was assigned a significantly higher **Risk Class 4** with a premium of \$17,000.00. This reinforces that the current applicant's profile is deemed substantially better than this highly similar historical case.

The ML model's classification of the current applicant into Risk Class 1, despite being a smoker, is noteworthy. It implies that other positive factors (like age if 23, healthy BMI, no claims, no family history, high income) are sufficiently strong to counteract the smoking risk, or that the comparable cases (especially APP-0642B5) also shared this characteristic.

3. Anomaly & Fraud Detection

* **Critical Anomaly - Age:** The reported age of **0.23 years** is the most significant anomaly. If it represents an infant, then the stated income (\$305,000.00), occupation ('WORKER'), and high coverage amount (\$4,500,000.00) are highly inconsistent and raise serious flags for potential data error or misrepresentation. This single point requires immediate verification. Assuming it's a typo for 23 years, the profile becomes more coherent.

* **Coverage Amount vs. Income:** A coverage amount of \$4,500,000.00 for an income of \$305,000.00 represents a multiple of approximately 14.75x. While high, this can be justifiable for high-income individuals with significant financial obligations (e.g., mortgages, business interests, dependents). It's at the higher end of typical income multiples and warrants a standard financial needs assessment to ensure alignment.

* **Smoking Status and Risk Class 1:** While not an anomaly per se, an applicant who smokes being assigned the best possible Risk Class 1 is somewhat uncommon and warrants scrutiny. It suggests either the model has found other overwhelming positive factors, or its training data includes scenarios where smokers, under specific conditions, can achieve this rating, or the smoking impact is being offset by a favorable young age (if 23).

* **Vague Occupation:** The generic 'WORKER' occupation, while not necessarily anomalous, lacks specificity, making it difficult to fully assess occupational risk.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 appears to be **justified based on the predicted Risk Class 1 and supporting historical data**, assuming the underlying data points are accurate.

* The machine learning model has assigned the lowest possible risk class (1), which naturally correlates with lower premiums.

* The prediction confidence of 60.1% is moderate. While not extremely high, it indicates a reasonable level of certainty in the model's assessment for this classification.

* The most similar historical case for risk class, APP-0642B5, also at Risk Class 1, was approved with the exact same premium of \$10,000.00. This provides strong internal validation for the proposed premium.

* Compared to other highly similar cases (POL-8902 at Risk Class 2, \$12,000 premium), the current applicant's lower risk class and premium are consistent.

However, the premium justification is **contingent on the resolution of the 'age' anomaly** and understanding how the smoking status was factored into the Risk Class 1 assignment. If the age is genuinely 0.23 years, or if the model's assessment of a smoker in Risk Class 1 is flawed without additional mitigating factors (e.g., exceptional health metrics not explicitly listed), then the premium might be insufficient.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review (Conditional Approval)****

Given the critical data discrepancy and specific risk factors, the application should not be automatically approved at this stage.

1. ****Immediate Action Required:**** ****Verify the applicant's age.**** This is paramount. If the age is confirmed as 23 years, the application can proceed to a detailed manual review. If it is genuinely 0.23 years, the application should be rejected immediately due to misrepresentation/incoherence, or referred for deep investigation of circumstances.
2. ****Manual Review Focus (assuming age 23):****
 - * ****Smoking Status:**** Reconfirm smoking status and duration. Investigate any specific health parameters that might allow a smoker to achieve Risk Class 1 according to underwriting guidelines.
 - * ****Occupation:**** Obtain a more specific occupation and industry to assess occupational hazard.
 - * ****Financial Underwriting:**** Conduct a thorough financial needs analysis to justify the high coverage amount relative to income, confirming it aligns with standard income multiples and the applicant's financial situation.
 - * ****ML Model Output Review:**** Underwriters should critically review the features that led to a Risk Class 1 for a smoker, especially given the moderate confidence score.

Subject to satisfactory clarification and verification of the age, and a favorable outcome from the focused manual review, ****conditional approval**** at the calculated premium of \$10,000.00 could be considered. Without clarification, the application should not proceed.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.